

網頁設計乙級

技術士技能檢定 · Flexible Study Plan

L440 · XAMPP · 3 sessions / week minimum

學科 EXAM 07/05 14:00–15:40 術科 EXAM Aug – Sep

STRATEGY IN ONE LINE Phone-drill 學科, train PHP/MySQL patterns on the L440, repeat the four fixed 術科 題組 until patterns become automatic. Missing days is normal — the plan is built for it.

01 Exam Facts

Working assumptions for the 115 年第 2 梯次 schedule, sourced from the official 技術士技能檢定 brochure and 網頁設計乙級術科應檢參考資料.

學科 date is 2026 / 07 / 05. For 網頁設計乙級, the time slot is afternoon 14:00–15:40. Practice your final mocks at 14:00 sharp.

術科 uses pre-announced questions: 4 題組 in total, each candidate completes one assigned 題組 on exam day.

術科 date is normally notified later by registered mail. Plan for late August through September runway.

Pressure release: 學科 and 術科 are registered together. Per the brochure, you may still take 術科 regardless of your 學科 score. Your 術科 prep is not wasted if July 5 does not go well.

Daily time is short, so this plan prioritizes repetition and pass probability over broad learning.

Missing days is expected. The plan uses weekly quotas, not daily streaks.

Priority order

Priority	Goal
1	Pass 學科 on July 5.
2	Build L440 / XAMPP habits early. 3 sessions / week minimum.
3	After July 5, shift to weekend timed 術科 builds.

02 Flexible Weekly System

★ UPDATED FOR FLEXIBLE SCHEDULE

Use a minimum weekly quota instead of a daily streak. Missing one or two days does not break the plan — only the week total matters. Do this every week. Time is short so the quota is fixed.

Week Modes

Week Mode	Sessions Required	What to do
Ideal week	3 short weekday + 1 weekend	Full Session A + B + C + phone 學科 daily
Normal week	2 short weekday + 1 weekend	Session B + C minimum; phone 學科 when possible
Emergency week	1 weekday + 1 weekend	Weekend rescue only — see below

Session Types

Session	Length	Action	Purpose
A — Setup / Foundation	30–45 min	XAMPP, localhost, phpMyAdmin, db.php, folder structure	Build exam-room reflex
B — PHP / MySQL Drill	30–45 min	One pattern only: SELECT / INSERT / UPDATE / DELETE / DELETE / INSERT	Build exam-room reflex
C — Weekend Build	1–3 hr	Notes app → 題組 1 section → 題組 2 → timed mock (late Swales)	Swales practice
Phone 學科	Flexible	Yamol / onlinetest / bquiz drilling — 5–10 questions while waiting	Pattern recognition

HARD RULE — Do not copy-paste during practice. Type, run, fix, delete, type again. The exam rewards recall and speed, not elegant code.

Miss Days? Use These Rules

Situation	Action
Miss 1 day	Do nothing special. Continue next session.
Miss 2–3 days	Skip typing warmup. Do the core task only (e.g. just db.php from memory).
Miss almost whole week	Weekend rescue: 30 min setup/fix · 30 min one PHP pattern · 30 min 學科 mock or wrong answers.

Weekly Tracker — fill in every Sunday

Must Finish (4 items = successful week)	Bonus
One L440 setup / practical session One PHP/MySQL pattern drill session One weekend build / review session One 學科 mock OR 50 phone questions	Extra typing drill: \$, _, ->, =>, [], {}, ; Extra 題組 practice section Wrong-answer notebook update

03 學科 Syllabus · Phone-Drill List

Drill these on the phone during dead time. Do not consume L440 desk time on 學科.

Topic	What to Drill
HTML	Tags, form elements, image / link paths, table and form basics
CSS	Selectors, priority, box model, display, positioning, fonts, colors
JavaScript/jQuery	Events, variables, conditions, form validation, DOM basics
PHP basics	Syntax recognition, GET / POST, session / cookie, include / require
MySQL / database	Database / table / field concepts, SELECT / INSERT / UPDATE / DELETE
Network / web concepts	URL, domain, browser, server, client, HTTP basics
Software / files	File extensions, folders, compression, image formats
Information security	Password, backup, malware, phishing, permissions
著作權法	Article-style questions, copyright scope, reproduction, public transmission, fair use keywords
個人資料保護法	Article-style questions, personal data definition, collection / processing / use, consent
共同科目	職業安全、工作倫理、環境保護、性別平等

學科 SCORE TARGET Before July 5, your mock score should hit 80–85+ repeatedly. Do not aim for 60 in practice — multi-select questions can drop your real score quickly.

04 術科 Survival Patterns

Files live under C:\xampp\htdocs\ . Browser path is http://localhost/foldername/. Every 題組 becomes manageable once you can build these patterns from memory.

4.1 Folder Structure

```
C:\xampp\htdocs\web-b\
project1\
index.php
admin\
index.php
add.php
edit.php
delete.php
includes\
db.php
css\ js\ images\
```

TIMED-RUN ROOT RULE Early learning can use nested folders like web-b/project1. But during timed mock runs, work directly under C:\xampp\htdocs\ so the browser path behaves like http://localhost/ and image paths do not surprise you on exam day.

4.2 db.php

PHP · includes/db.php

```
<?php
session_start();
$conn = mysqli_connect("localhost", "root", "", "db_name");
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
mysqli_set_charset($conn, "utf8");
?>
```

4.3 SELECT All Rows

PHP

```
$sql = "SELECT * FROM table_name";
$result = mysqli_query($conn, $sql);
while ($row = mysqli_fetch_assoc($result)) {
    echo $row['title'];
}
```

4.4 SELECT One Row with WHERE

Critical for edit.php and detail.php.

PHP

```
$id = $_GET['id'];
$sql = "SELECT * FROM table_name WHERE id='$id'";
$result = mysqli_query($conn, $sql);
$row = mysqli_fetch_assoc($result);
```

4.5 SELECT by Category / Type

PHP

```
$type = $_GET['type'];
$sql = "SELECT * FROM table_name WHERE type='$type'";
$result = mysqli_query($conn, $sql);
```

4.6 INSERT

PHP

```
$title = $_POST['title'];
$sql = "INSERT INTO table_name (`title`) VALUES ('$title')";
mysqli_query($conn, $sql);
```

If time permits — basic escaping variant for stricter security marking:

PHP · safer variant

```
$title = mysqli_real_escape_string($conn, $_POST['title']);
$sql = "INSERT INTO table_name (`title`) VALUES ('$title')";
mysqli_query($conn, $sql);
```

4.7 UPDATE

PHP

```
$id = $_GET['id'];
$title = $_POST['title'];
$sql = "UPDATE table_name SET `title`=' $title' WHERE `id`=' $id'";
mysqli_query($conn, $sql);
```

4.8 DELETE

PHP

```
$id = $_GET['id'];
$sql = "DELETE FROM table_name WHERE `id`=' $id'";
mysqli_query($conn, $sql);
```

4.9 Login / Session Pattern

PHP · session guard

```
session_start();
$_SESSION['login'] = true;
if (!isset($_SESSION['login'])) {
header("Location: login.php");
exit();
}
```

4.10 Image Upload

MEMORY TRIGGER — File upload requires BOTH: HTML form enctype="multipart/form-data" AND PHP \$_FILES handling. Forgetting enctype is the #1 way to break upload features under pressure.

HTML

```
<form method="post" enctype="multipart/form-data">
<input type="file" name="image">
<button type="submit">Upload</button>
</form>
```

PHP

```
$upload_dir = "../images/";
$filename = basename($_FILES["image"]["name"]);
move_uploaded_file(
$_FILES["image"]["tmp_name"],
$upload_dir . $filename
);
```

4.11 Image Display

Most 題組 require both saving AND displaying images. Path resolution trips people up: relative to the displaying page, with ../ when the page is inside admin/.

HTML + PHP

```
<!-- From a page at root, e.g. index.php -->


<!-- From a page inside admin/, e.g. admin/list.php -->

```

05 Timeline · May 10 onwards

★ UPDATED FOR FLEXIBLE SCHEDULE

May 10 to July 5 is 8 calendar weeks including exam week. Enough for 學科 + 術科 foundation. After July 5, shift hard into timed builds. Count weekly deliverables, not daily tasks — missing a day does not break the week.

Wk	Dates	Focus	Deliverable
W0	May 10–16	Setup + baseline	L440, XAMPP, VS Code, localhost / phpMyAdmin. One 學科 baseline mock. Type db.php from mem
W1	May 17–23	CRUD foundation	SELECT all, SELECT WHERE, INSERT, UPDATE, DELETE. Build a tiny notes admin.
W2	May 24–30	Login + upload + display	Session / login / logout. Image upload form + \$_FILES. Image display in list page.
W3	May 31–Jun 6	題組 1 first pass	卓越科技大學. Folder, DB, frontend skeleton, admin login, add / edit / delete. No speed target.
W4	Jun 7–13	題組 1 repeat + 題組 2	Repeat 題組 1 core. Start 健康促進網. Phone drill 學科 daily.
W5	Jun 14–20	題組 2 + law drill	題組 2 admin / display. Add explicit 著作權法 / 個資法 phone drilling.
W6	Jun 21–27	題組 3 / 4 survival	ABC 影城 + 精品電子商務 structure. Identify must-do vs skip-if-stuck.
W7	Jun 28–Jul 4	學科 final week	Daily mock + wrong-answer review. Two mocks at 14:00 sharp. Light practical only.
—	Jul 5	學科 exam day	14:00–15:40. Light review only. Your 術科 continues regardless.
W8	Jul 6–12	Post-學科 + 題組 1 / 2	One 題組 1 timed run, one 題組 2 timed / semi-timed run.
W9	Jul 13–19	題組 3 / 4 survival	High-value features only. Build must-do / should-do / skip lists.
W10	Jul 20–26	Cheatsheet + guided	Create cheatsheet.php. Mocks may use it, but only after trying from memory.
W11	Jul 27–Aug 2	Wean off cheat sheet	Open cheatsheet only when stuck. Random 題組 partial / full runs.
W12	Aug 3–9	No-cheat timed week	Wi-Fi off, timer on, official PDF only. At least one full 3–4 hour run.
—	Aug 10–31	Maintenance	Rotate 題組. Focus on error log + WHERE / upload / session.
—	No notice by Aug 31	Soft cap	Reduce to 2 sessions / week. Restart full runs after notice.

06 Compression Rules if Start Date Slips

Situation	Adjustment
Start May 10	Use the standard schedule above.
Start after June 1	Collapse CRUD + login + upload into one intense week. Skip broad PHP learning.
Start after mid-June	Go directly: db.php → CRUD → SELECT WHERE → upload → 題組 1 → 題組 2 → 3 / 4 survival.
術科 notice arrives early	Stop expanding theory. Run timed mocks every weekend. Drill only your top 10 recurring errors.

07 Weekly Task Detail Before July 5

W0 · May 10–16 · Setup + Baseline

Check L440 hardware and keyboard.

Install XAMPP, VS Code, Chrome / Edge, CrystalDiskInfo.

Confirm http://localhost and phpMyAdmin both work.

Run one 學科 mock and record the score.

Type db.php from memory twice.

W1 · May 17–23 · CRUD Foundation

Mon — SELECT all

Tue — SELECT WHERE one row

Wed — INSERT form

Thu — UPDATE edit.php

Fri — DELETE

Weekend — tiny notes admin app, fully working.

W2 · May 24–30 · Login + Upload + Display

Build login.php / logout.php / admin guard.

Type enctype="multipart/form-data" 10 times.

Practice move_uploaded_file.

Practice from index, and from admin/.

Weekend — notes app with image upload + display.

W3 · May 31–Jun 6 · 題組 1 First Pass

Read 題組 1 requirements end to end.

Build folder and database.

Build admin login + one CRUD section.

Don't chase perfection. Get it running.

W4 · Jun 7–13 · 題組 1 Repeat + 題組 2 Start

Repeat 題組 1 database / admin patterns.

Start 題組 2 skeleton.

Mark which functions are shared between 題組 1 and 2.

W5 · Jun 14–20 · 題組 2 + Law Drill

題組 2 main admin / display features.

Daily phone drill: 著作權法 + 個資法 questions.

Weekend — 題組 2 semi-timed run.

W6 · Jun 21–27 · 題組 3 / 4 Survival Intro

Build only high-value parts of each.

Create skip-if-stuck list for ABC 影城 and 精品電子商務.

Keep 學科 mock score above 80.

W7 · Jun 28–Jul 4 · Final 學科 Week

One mock every 1–2 days.

Two mocks at 14:00 sharp to match exam-day timing.

Review wrong-answer notebook.

Light practical only — db.php, CRUD, upload, session.

Sleep normally before exam.

08 July 5 · Exam Morning

BRING THESE OR YOU CANNOT SIT THE EXAM

Item	
准考證 (admission ticket)	
身分證 (national ID)	Required for identity check
2B pencil	For any paper-based marking
Black or blue pen	
Watch (no smartwatch)	Smartwatches will be removed

Morning Routine

Wake at your normal time. Eat a normal breakfast — not a special one.

Light review only — your wrong-answer notebook, max 30 minutes.

No new material. Nothing you have not already practiced.

Leave so you arrive at the exam venue 30 minutes early.

Confirm you have all required documents before leaving home.

AFTER THE EXAM Do not score yourself from memory or check answer keys obsessively. Even if it feels rough, your 術科 path is unaffected. Take the rest of July 5 off. Resume W8 on July 6.

09 Post-學科 Practical Plan

Period	Main Task	Rule
Jul 6–12	題組 1 + 2 timed / semi-timed runs	Write every error to mistake log.
Jul 13–19	題組 3 + 4 survival runs	High-value features first. Don't get trapped on one feature.
Jul 20–26	Create cheatsheet.php	Training wheel, not a crutch.
Jul 27–Aug 2	Random 題組 runs	Open cheatsheet only after 5 minutes stuck.
Aug 3–9	No-cheat full mock	Official PDF only. Wi-Fi off. Timer on.
Aug 10–31	Maintenance loop	Repeat weakest 題組 + top recurring errors.
No notice by Aug 31	Soft cap	Reduce to 2 sessions / week.

10 題組 Priority + Survival Strategy

#	題組	Goal
1	卓越科技大學校園資訊系統	Master strongly. Foundation for admin, DB, CRUD, image handling.
2	健康促進網社群平台	Master strongly. Shared backend / admin logic, likely good scoring base.
3	ABC 影城	Survival. Movie display + admin + basic booking flow.
4	精品電子商務	Survival. Products, admin management, cart / order basics, upload.

EXAM-ROOM TRIAGE — Never spend 40 minutes on one broken feature. Secure DB connection, page display, admin CRUD, session, and uploads first. Skip low-value extras when stuck.

11 cheatsheet.php · Build by Week 10

Save at C:\xampp\htdocs\cheatsheet.php. Week 10 — usable during mocks. Week 11 — only when stuck. Week 12 — closed.

PHP · cheatsheet.php

```
<?php
// ===== db.php =====
session_start();
$conn = mysqli_connect("localhost", "root", "", "db_name");
mysqli_set_charset($conn, "utf8");

// ===== SELECT all =====
$sql = "SELECT * FROM table_name";
$result = mysqli_query($conn, $sql);

// ===== SELECT one =====
$id = $_GET['id'];
$sql = "SELECT * FROM table_name WHERE id='$id'";
$result = mysqli_query($conn, $sql);
$row = mysqli_fetch_assoc($result);

// ===== INSERT =====
$title = $_POST['title'];
// Safer: $title = mysqli_real_escape_string($conn, $_POST['title']);
$sql = "INSERT INTO table_name (`title`) VALUES ('$title')";
mysqli_query($conn, $sql);

// ===== UPDATE =====
$id = $_GET['id'];
$title = $_POST['title'];
// Safer: $title = mysqli_real_escape_string($conn, $_POST['title']);
$sql = "UPDATE table_name SET `title`='$title' WHERE `id`='$id'";
mysqli_query($conn, $sql);

// ===== DELETE =====
$id = $_GET['id'];
$sql = "DELETE FROM table_name WHERE `id`='$id'";
mysqli_query($conn, $sql);

// ===== SESSION guard =====
if (!isset($_SESSION['login'])) {
    header("Location: login.php");
    exit();
}

// ===== IMAGE UPLOAD =====
// HTML form MUST have enctype="multipart/form-data"
$upload_dir = "../images/";
$filename = basename($_FILES["image"]["name"]);
move_uploaded_file($_FILES["image"]["tmp_name"], $upload_dir . $filename);

// ===== IMAGE DISPLAY =====
// From root page: 
// From admin/ page: 
?>
```

12 Common Error Checklist

When something breaks, scan this list before suspecting your logic.

Common Error	Fix / Memory Cue
Apache / MySQL not started	First 30 seconds — XAMPP green checks.
File saved outside htdocs	Always check C:\xampp\htdocs\projectname.
Wrong URL	Folder name maps to http://localhost/foldername/.
Forgot semicolon ;	Scan the line above the error.
Forgot \$ in variable	PHP variables always use \$.
Wrong table / column name	Compare with phpMyAdmin exactly.
Wrong form input name	input name must match \$_POST key.
Forgot WHERE in UPDATE/DELETE	Without WHERE, you affect all rows.
Forgot session_start()	Session must start before \$_SESSION.
Forgot utf8 charset	mysqli_set_charset(\$conn, "utf8").
Upload broken	Check enctype="multipart/form-data" first.
Image not showing	Check ../images/ vs images/ — depends on page location.
header() error	No output before header() — no echo, no whitespace before <?php.

13 First 7 L440 Sessions

Complete these in order across your first week — not necessarily one per day.

Sess.	Task
S1	Install / check tools — XAMPP, VS Code, Chrome, Edge. Open localhost + phpMyAdmin.
S2	Create C:\xampp\htdocs\web-b\test1 for learning. (Timed runs later use htdocs root directly.) Create index.php and includes/db.php.
S3	Create database in phpMyAdmin. Connect db.php. Echo "Connected".
S4	Create one table. Insert sample data manually. SELECT and display it.
S5	Create add.php with HTML form + INSERT.
S6	Create edit.php using SELECT WHERE + UPDATE.
S7	Create delete.php. Run one 學科 mock. Write mistake log.

14 Weekly Scorecard

Fill in every Sunday. Don't trust your feeling — trust the numbers.

Metric	Target before Jul 5	Target before 術科
Typing PHP symbols	5 / 10	7 / 10
db.php from memory	8 / 10	9 / 10
SELECT + SELECT WHERE	7 / 10	9 / 10
INSERT / UPDATE / DELETE	7 / 10	9 / 10
Session / login	6 / 10	8 / 10
Image upload	6 / 10	8 / 10
Image display + path	6 / 10	8 / 10
題組 1	First pass	8 / 10
題組 2	First pass	8 / 10
題組 3	Recognize structure	5–6 / 10 survival
題組 4	Recognize structure	5–6 / 10 survival
學科 mock score	80–85+	Already passed

15 References

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著作權法 — law.moj.gov.tw

個人資料保護法 — law.moj.gov.tw

FINAL REMINDER This plan is built for real life: 3 sessions a week is enough. Missing days is not failure. Complete the weekly quota, drill the exact patterns, repeat them on the L440, use phone time for 學科 repetition. Boring repetition is the strategy.